

Ronald J. Tackett, Ph.D.

Founding Head of School & Associate Professor of Physics

School of Foundational Studies · Kettering University



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EDUCATION

Ph.D., Condensed Matter Physics 2003–2008

Wayne State University, Detroit, MI

- Dissertation: *The Synthesis, Characterization, and Application of Multifunctional Magnetic Nanoparticles*
- Advisor: Dr. Gavin Lawes

B.S., Engineering Physics 2001–2003

Eastern Michigan University, Ypsilanti, MI

PROFESSIONAL EXPERIENCE

Founding Head (Dean-Equivalent), School of Foundational Studies 2025–pres.

Kettering University, Flint, MI

Adjunct (Volunteer) Associate Professor of Physics 2025–pres.

Wayne State University, Detroit, MI

Associate Professor of Physics 2019–pres.

Kettering University, Flint, MI

Assistant Professor of Physics 2014–2019

Kettering University, Flint, MI

Assistant Professor of Physics 2010–2014

Arkansas Tech University, Russellville, AR

Postdoctoral Research Associate 2008–2010

University of Texas at El Paso, El Paso, TX

Graduate Research Assistant 2004–2008

Wayne State University, Detroit, MI

Graduate Teaching Assistant 2003–2004

Wayne State University, Detroit, MI

PUBLICATIONS

- B.R. Dahal, K. Schroeder, M.M. Allyn, **R.J. Tackett**, Y. Huh, and P. Kharel. "Near-room-temperature magnetocaloric properties of $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$ ($x = 0.11, 0.17$, and 0.19) nanoparticles." *Mat. Res. Express* **5**, 106103 (2018).
- R.J. Tackett**, J. Thakur, N. Mosher, E. Perkins-Harbin, R.E. Kumon, L. Wang, C. Rablau, and P.P. Vaishnava. "A method for measuring the Néel relaxation time in a frozen ferrofluid." *J. Appl. Phys.* **118**(6), 064701 (2015).
- Y.C.N. Chen, C.Y. Hsieh, **R. Tackett**, P. Kokeny, R.K. Regmi, and G. Lawes. "Magnetic moment quantifications of small spherical objects in MRI." *Mag. Res. Imaging* **33**(6), 829 (2015).
- C.E. Botez, A.H. Adair, and **R.J. Tackett**. "Evidence of superspin-glass behavior in $\text{Zn}_{0.5}\text{Ni}_{0.5}\text{Fe}_2\text{O}_4$ nanoparticles." *J. Phys. Cond. Mat.* **27**(7), 076005 (2015).
- S.S. Laha, **R.J. Tackett**, and G. Lawes. "Interactions in $\gamma\text{-Fe}_2\text{O}_3$ and Fe_3O_4 nanoparticle systems." *Physica B* **448**, 69 (2014).
- C.E. Botez, **R.J. Tackett**, J.D. Hermosillo, J.Z. Zhang, Y.S. Zho, and L.P. Wang. "High-pressure synchrotron x-ray diffraction studies of superprotonic transitions in phosphate solid acids." *Solid State Ionics* **213**, 58 (2012).
- R. Tackett**, G. Lawes, R. Suryanarayanan, M. Apostu, and A. Revcolevschi. "The zero-field glassy ground state and field-induced ferromagnetic transition in $(\text{La}_{0.4}\text{Pr}_{0.6})_{1.2}\text{Sr}_{1.8}\text{Mn}_2\text{O}_7$." *J. Phys. Cond. Mat.* **23**, 156004 (2011).
- K.K. Bharathi, **R.J. Tackett**, C.E. Botez, and C.V. Ramana. "Coexistence of spin glass behavior and long range ferromagnetic ordering in La- and Hy-doped Co ferrite." *J. Appl. Phys.* **109**, 07A510 (2011).
- C.E. Botez, A.W. Bhuiya, and **R.J. Tackett**. "Dynamic-susceptibility studies of the interplay between the Néel and Brown magnetic relaxation mechanisms." *App. Phys. A* **104**, 177 (2011).
- C.E. Botez, D. Carbajal, K. Adiraju, and **R.J. Tackett**. "Intermediate-temperature polymorphic phase transition in KH_2PO_4 : A synchrotron x-ray diffraction study." *J. Phys. Chem. Sol.* **71**, 1576 (2010).
- R.J. Tackett**, J.G. Parsons, B.I. Machado, S. Gaytan, L.E. Murr, and C.E. Botez. "Evidence of low-temperature superparamagnetism in Mn_3O_4 nanoparticle ensembles." *Nanotechnology* **21**, 265703 (2010).
- R.J. Tackett**, A.W. Bhuiya, and C.E. Botez. "Dynamic susceptibility evidence of surface spin freezing in ultrafine NiFe_2O_4 nanoparticles." *Nanotechnology* **20**, 445705 (2009).
- C.E. Botez, H. Martinez, **R.J. Tackett**, R.R. Chianelli, J. Zhang, and Y. Zhao. "High-temperature crystal structures and chemical modifications in RbH_2PO_4 ." *J. Phys. Cond. Mat.* **21**, 325401 (2009).
- K. Senevirathne, **R. Tackett**, P.R. Kharel, G. Lawes, K. Somaskandan, and S.L. Brock. "Discrete, dispersible MnAs nanocrystals from solution methods: phase control on the nanoscale and magnetic consequences." *ACS Nano* **3**, 1129 (2009).
- R. Regmi, **R. Tackett**, and G. Lawes. "Suppression of low-temperature magnetic states in Mn_3O_4 nanoparticles." *J. Magn. Magn. Mater.* **321**, 2296 (2009).
- B.C. Melot, **R. Tackett**, J. O'Brien, A.L. Hector, G. Lawes, R. Seshadri, and A.P. Ramirez. "Large low-temperature specific heat in pyrochlore $\text{Bi}_2\text{Ti}_2\text{O}_7$." *Phys. Rev. B* **79**, 224111 (2009).
- S. Putatunda, A. Singer, **R. Tackett**, and G. Lawes. "Development of a high-strength, high-toughness ausferritic steel." *Metall. Mater. Trans. A* **513–514**, 329 (2009).
- Y. Shen, Y.C.N. Cheng, G. Lawes, J. Neelavalli, C. Sudakar, **R. Tackett**, H.P. Ramanth, and E.M. Haacke. "Quantifying magnetic nanoparticles in non-steady flow by MRI." *Magn. Reson. Mater. Phys.* **21**, 345 (2008).
- C. Rablau, P.P. Vaishnava, C. Sudakar, **R. Tackett**, G. Lawes, and R. Naik. "Magnetic field induced optical anisotropy in ferrofluids: A time dependent light scattering investigation." *Phys. Rev. E* **78**, 051502 (2008).
- R. Tackett**, C. Sudakar, R. Naik, G. Lawes, C. Rablau, and P.P. Vaishnava. "Magnetic and optical response of tuning the magnetocrystalline anisotropy in Fe_3O_4 ferrofluids by Co doping." *J. Magn. Magn. Mater.* **320**, 2755 (2008).
- S. Singh, R. Suryanarayanan, **R. Tackett**, G. Lawes, A.K. Sood, P. Berthet, and A. Revcolevschi. "Spin-ice state in the geometrically frustrated metallic-ferromagnet $\text{Sm}_2\text{Mo}_2\text{O}_7$." *Phys. Rev. B* **77**, 054408 (2008).
- P.P. Vaishnava, **R. Tackett**, A. Dixit, C. Sudakar, R. Naik, and G. Lawes. "Magnetic relaxation and dissipative

- heating in ferrofluids." *J. Appl. Phys.* **102**, 063914 (2007).
- R. Tackett**, G. Lawes, B.C. Melot, M. Grossman, E.S. Toberer, and G. Lawes. "Magnetodielectric coupling in Mn_3O_4 ." *Phys. Rev. B* **76**, 024409 (2007).
- J. Cao, R.C. Rai, S. Brown, J. Musfeldt, **R. Tackett**, G. Lawes, Y. Wang, X. Wei, M. Apostu, R. Suryanarayanan, and A. Revcolevschi. "Observation of 300 K high energy magnetodielectric response in the bilayer manganite $(\text{La}_{0.4}\text{Pr}_{0.6})_{1.2}\text{Sr}_{1.8}\text{Mn}_2\text{O}_7$." *Appl. Phys. Lett.* **91**, 021913 (2007).
- S.K. Putatunda, S. Kesani, **R. Tackett**, and G. Lawes. "Development of austenite-free ADI (austempered ductile cast iron)." *Mater. Sci. Eng. A* **435–436**, 112 (2006).
- G. Lawes, **R. Tackett**, O. Masala, B. Adhikary, R. Naik, and R. Seshadri. "Positive and negative magnetocapacitance in magnetic nanoparticle systems." *Appl. Phys. Lett.* **88**, 242903 (2006).
- G. Tsoi, U. Seneratne, **R. Tackett**, E. Buc, R. Naik, P. Vaishnava, V. Naik, and L. Wenger. "Memory effects and magnetic interactions in $\gamma\text{-Fe}_2\text{O}_3$ nanoparticle system." *J. Appl. Phys.* **97(10)**, 10J507 (2005).
- G. Tsoi, L. Wenger, U. Seneratne, **R. Tackett**, E. Buc, R. Naik, P. Vaishnava, and V. Naik. "Memory effects in a superparamagnetic $\gamma\text{-Fe}_2\text{O}_3$ system." *Phys. Rev. B* **72**, 014445 (2005).

AWARDS, FELLOWSHIPS, & GRANTS

Distinguished Service Award <i>Kettering University</i>	2025
Strategic Investment Grant (Co-PI) <i>Michigan Economic Development Corporation</i>	2024
Outstanding New Research Award <i>Kettering University</i>	2018
Provost's Research Matching Fund <i>Kettering University</i>	2017
Faculty Research Fellowship <i>Kettering University</i>	2016
Provost's Matching Fund <i>Kettering University</i>	2016
Provost's Travel Fund <i>Kettering University</i>	2016
Provost's Travel Fund <i>Kettering University</i>	2015
Major Research Instrumentation (1531402) <i>National Science Foundation</i>	2015
Faculty Research Fellowship <i>Kettering University</i>	2014
Undergraduate Research Award <i>Arkansas Tech University</i>	2013
Junior Science and Humanities Symposium <i>The Academy of Applied Science</i>	2013
Arkansas Space Grant Symposium <i>NASA</i>	2012
Junior Science and Humanities Symposium <i>The Academy of Applied Science</i>	2012
Arkansas Space Grant Symposium <i>NASA</i>	2011

PRESENTATIONS

Invited Talks

2026. *In the Age of AI, What Are We Actually Educating Students For?* Second Gathering, Council of Liberal Arts Leaders at Technological Universities (CoLLT), Georgia Institute of Technology, Atlanta, GA.
2026. *The School of Foundational Studies at Kettering University: An Institutional Profile*. Pre-Planning Meeting, Council of Liberal Arts Leaders at Technological Universities (CoLLT), Georgia Institute of Technology, Atlanta, GA.
2024. *A Journey Through Innovation and Impact: How Semiconductors Revolutionized the World*. Longway Planetarium, Flint, MI.
2023. *Through the Looking Glass: A Discussion of the Art of Microscopy*. Longway Planetarium, Flint, MI.
2019. *Specialization and Optimization of Metal-Oxide Nanoparticles for Use in the Magnetic Hyperthermia Treatment of Cancer*. Fall Meeting, Midland Section, American Chemical Society, Saginaw Valley State University, Saginaw, MI.
2018. *Tuesday Teaching Talk: EndNote*. Center for Excellence in Teaching and Learning, Kettering University, Flint, MI.
2017. *Through the Magnetic Lens: A Discussion of Electron Microscopy*. Science on Tap, Tenacity Brewing Company, Flint, MI.
2016. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles*. Gavin Lawes Memorial Symposium, Wayne State University, Detroit, MI.
2016. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles*. SPS Guest Lecture Series, Lawrence Technological University, Southfield, MI.
2015. *Treating Big Problems with Small Things: Applications of Magnetic Nanoparticles*. Distinguished Faculty Series, Kettering University, Flint, MI.
2014. *Magnetic Fluid Hyperthermia Treatment of Cancer using Dextran-Coated Fe₃O₄ Nanoparticles*. Solid State Physics Seminar, Wayne State University, Detroit, MI.
2009. *It's a Small World After All: Characterization and Applications of Magnetic Nanoparticles*. The University of Texas at El Paso, El Paso, TX.
2009. *High Temperature Phase Transitions in CsH₂PO₄ and RbH₂PO₄*. New Mexico State University, Las Cruces, NM.
2006. *The Synthesis and Characterization of Magnetic Nanoparticles*. Université Paris-Sud, Orsay, France.

Contributed Presentations

- J. Wylie and **R.J. Tackett**. 2019. Synthesis and Characterization of Gadolinium-Doped Magnetite. Fall Meeting, Eastern Great Lakes Section, APS, Kettering University, Flint, MI.
- J. Wylie and **R.J. Tackett**. 2019. Synthesis and Characterization of Gadolinium-Doped Magnetite. Michigan Academy of Science, Arts, and Letters (MASAL), Alma College, Alma, MI.
- M. Warteman and **R.J. Tackett**. 2019. Effects of Doping Concentration in the Sol-Gel Synthesis of Strontium-Doped Lanthanum Manganate. MASAL, Alma College, Alma, MI.
- R.J. Tackett**, M.M. Allyn, V.K. Garg, A.C. de Oliveira, and P.P. Vaishnava. 2017. A Comparison of Methods for the Determination of the Magnetocrystalline Anisotropy Constant. Spring Meeting, Ohio Section, APS, Eastern Michigan University, Ypsilanti, MI.
- M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2017. The Investigation of Smart Magnetic Nanoparti-

- cles for Use in the Hyperthermia Treatment of Cancer. Spring Meeting, Ohio Section, APS, Eastern Michigan University, Ypsilanti, MI.
- R.J. Tackett**, M.M. Allyn, V.K. Garg, A.C. de Oliveira, E.A. Elhamid, and P.P. Vaishnava. 2016. A Comparison of Methods for the Determination of the Magnetocrystalline Anisotropy Constant. March Meeting, APS, Baltimore, MD.
- M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2016. The Investigation of Smart Magnetic Nanoparticles for Use in the Hyperthermia Treatment of Cancer. March Meeting, APS, Baltimore, MD.
- M.M. Allyn, P. Kharel, P.P. Vaishnava, and **R.J. Tackett**. 2016. The Investigation of Smart Magnetic Nanoparticles for Use in the Hyperthermia Treatment of Cancer. MASAL, Saginaw Valley State University, Saginaw, MI.
- N. Mosher, E. Perkins-Harbin, B. Aho, L. Wang, R.E. Kumon, C. Rablau, P.P. Vaishnava, and **R.J. Tackett**. 2015. Determination of the Magnetocrystalline Anisotropy Constant from Frequency-Dependence of the Specific Absorption Rate in a Frozen Ferrofluid. March Meeting, APS, San Antonio, TX.
- J.S. Pennington and **R.J. Tackett**. 2013. Variant Crystal Lattice Structure of $\gamma\text{-Fe}_2\text{O}_3$ Under Successive Loadings Into an Alginate Polymer Matrix. Arkansas Space Grant Consortium, Morrilton, AR; Arkansas Academy of Science, Little Rock, AR.
- J.D. Counts and **R.J. Tackett**. 2013. Particle Size Dependence of Superparamagnetic Blocking in Magnetite (Fe_3O_4) Nanoparticles. Arkansas Space Grant Consortium, Morrilton, AR.
- J.D. Counts and **R.J. Tackett**. 2012. Particle Size Dependence of Superparamagnetic Blocking in Fe_3O_4 Nanoparticles. Arkansas Academy of Science, Southern Arkansas University, Magnolia, AR; Senior Honors Symposium, Arkansas Tech University, Russellville, AR.
- J.C. Smith and **R.J. Tackett**. 2011. Simulation of Powder X-Ray Diffraction Using a Polycrystalline Film of Spherical Colloids. Arkansas Academy of Science, University of Arkansas at Monticello, AR.
- R.J. Tackett**, H. Martinez, R.R. Chianelli, J. Zhang, Y. Zhao, and C.E. Botez. 2010. High-Temperature Crystal Structures and Chemical Modifications in RbH_2PO_4 . March Meeting, APS, Portland, OR.
- C. Rablau, P.P. Vaishnava, R. Naik, G. Lawes, **R. Tackett**, and C. Sudakar. 2008. Tuning of the Magnetocrystalline Anisotropy in $\text{Co}_x\text{Fe}_{3-x}\text{O}_4$ Nanoparticles. March Meeting, APS, New Orleans, LA.
- R. Tackett**, C. Sudakar, R. Naik, G. Lawes, C. Rablau, and P.P. Vaishnava. 2008. March Meeting, APS, New Orleans, LA.
- R. Tackett**, E. Toberer, R. Seshadri, and G. Lawes. 2007. Magnetodielectric Coupling in Mn_3O_4 . March Meeting, APS, Denver, CO.
- R. Tackett**, O. Masala, B. Adhikary, R. Naik, A.P. Ramirez, R. Seshadri, and G. Lawes. 2006. The Coupling of Magnetic and Dielectric Properties in Magnetic Nanoparticles. March Meeting, APS, Los Angeles, CA.

TEACHING EXPERIENCE

Kettering University

Introduction to Semiconductors <i>Lead Instructor</i>	SCI-100
Introduction to Material Science & Engineering <i>Lead Instructor</i>	EP-342
Solid State Physics <i>Lead Instructor</i>	EP-446
College Mathematics <i>Lead Instructor</i>	MATH-100
Newtonian Mechanics <i>Lead Instructor</i>	PHYS-114

Newtonian Mechanics Laboratory <i>Lead Instructor</i>	PHYS-115
Electricity & Magnetism <i>Lead Instructor</i>	PHYS-224
Electricity & Magnetism Laboratory <i>Lead Instructor</i>	PHYS-225
The Science of Sensors <i>Lead Instructor</i>	PHYS-304
Thermodynamics & Statistical Physics <i>Lead Instructor</i>	PHYS-452
Quantum Mechanics <i>Lead Instructor</i>	PHYS-462

Arkansas Tech University

Physical Science Laboratory <i>Lead Instructor</i>	PHSC-1021
Introduction to Physical Science <i>Lead Instructor</i>	PHSC-1013
Physics Laboratory I <i>Lead Instructor</i>	PHYS-2000
Physics Laboratory II <i>Lead Instructor</i>	PHYS-2010
Algebra-Based Physics I <i>Lead Instructor</i>	PHYS-2014
Algebra-Based Physics II <i>Lead Instructor</i>	PHYS-2024
Calculus-Based Physics I <i>Lead Instructor</i>	PHYS-2114
Calculus-Based Physics II <i>Lead Instructor</i>	PHYS-2124
Solid State Physics <i>Lead Instructor</i>	PHYS-3153
Advanced Physics Laboratory <i>Lead Instructor</i>	PHYS-4113

RESEARCH MENTORING

Bradley Henry <i>Senior Research Thesis, Kettering University</i>	2021
Joshua Wiley <i>Senior Research Thesis, Kettering University</i>	2020
Przemyslaw Piotrowski <i>Senior Research Thesis, Kettering University</i>	2019
Megan Allyn <i>Senior Research Thesis, Kettering University</i>	2018
Joy Counts <i>Undergraduate Research Assistant, Arkansas Tech University</i>	2013

SERVICE

Service to the Field

Treasurer, Eastern Great Lakes Section <i>American Physical Society</i>	2026–pres.
Past Chair, Eastern Great Lakes Section <i>American Physical Society</i>	2026–pres.
Founding Member <i>Council of Liberal Arts Leaders at Technological Universities</i>	2026–pres.
Chair, Eastern Great Lakes Section <i>American Physical Society</i>	2025–2026
Advocacy Champion <i>American Physical Society</i>	2025–pres.
Chair-Elect, Eastern Great Lakes Section <i>American Physical Society</i>	2024–2025
Vice Chair, Eastern Great Lakes Section <i>American Physical Society</i>	2023–2024
Review Panelist (P232162) <i>National Science Foundation</i>	2023
Reviewer <i>AIP Advances</i>	2021–pres.
Reviewer <i>Journal of Alloys and Compounds (Elsevier)</i>	2020–pres.
Reviewer <i>American Journal of Physics (AAPT)</i>	2018–pres.
Reviewer <i>NSF Graduate Research Fellowship Program</i>	2017–2020
Reviewer <i>The Physics Teacher (AAPT)</i>	2017–pres.
Reviewer <i>National Defense Science and Engineering Graduate Fellowship</i>	2016–2017

Service to the University

Reviewer, President's Medal Nominations <i>Kettering University</i>	2025
Founding Chair, Instructional Technology Advisory Committee <i>Kettering University</i>	2024–2025
Moderator, Faculty Senate <i>Kettering University</i>	2024–2025
Moderator-Elect, Faculty Senate <i>Kettering University</i>	2023–2024
Member, Employee Satisfaction and Fulfillment Working Group <i>Kettering University</i>	2022–2023

Senator (Natural Sciences), Faculty Senate <i>Kettering University</i>	2020–2023
Chair, Faculty Senate Thesis Committee <i>Kettering University</i>	2020–2023
Secretary, Faculty Senate <i>Kettering University</i>	2019–2020
Member, Advisory Board for Excellence in Teaching & Learning <i>Kettering University</i>	2019–2025
Member, University Curriculum Committee <i>Kettering University</i>	2016–2018

Service to the Community

Commencement Speaker <i>Livonia Stevenson High School</i>	2026
Volunteer (Pit/Props) <i>Grand Blanc High School Marching Band</i>	2025–pres.
Facilitator, Quantum Science Workshop <i>Genesee Career Institute</i>	2025
Secretary, Brendel Elementary PTO <i>Grand Blanc, MI</i>	2020–2023
Member, Zero-Mil Increase Bond Committee <i>Grand Blanc Community Schools</i>	2020
Presenter, Career Day <i>Brendel Elementary School, Grand Blanc, MI</i>	2019, 2020

Professional Society Membership

American Physical Society (APS) <i>Member</i>	2008–pres.
American Association of Physics Teachers (AAPT) <i>Member</i>	2015–pres.
American Association for the Advancement of Science (AAAS) <i>Member</i>	2019–pres.
Materials Research Society (MRS) <i>Member</i>	2019–2021
Advanced Laboratory Physics Association (ALPhA) <i>Member</i>	2019–pres.